

## OBJECTIVE

The main goal of Lab 4 is to see how rise time, settling time, and percentage overshoot change when the damping ratio has different values.

## METHODOLOGY & SOLUTIONS

### Task 3: Second-Order System Analysis

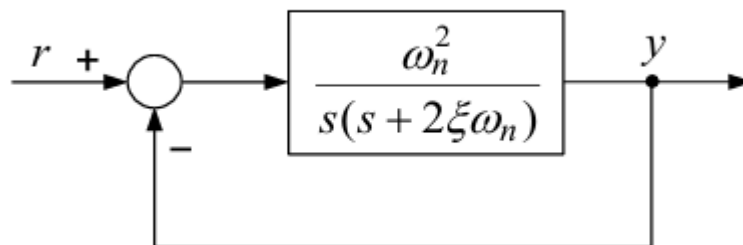
Rise time is the time a signal needs to go from a low level to a high level. It is usually measured from 10% to 90% of its final value. There is no exact single formula for rise time, but we can use this approximation:  $\frac{2.16\zeta + 0.60}{\omega_n}$

If the rise time is short, it means the system responds faster.

Settling time is the time needed for the system output to reach its final value and stay within a small error range around it (in our case,  $\pm 2\%$ ). It shows how stable the system is. For the 2% condition, settling time is calculated as:  $\frac{4}{\zeta\omega_n}$

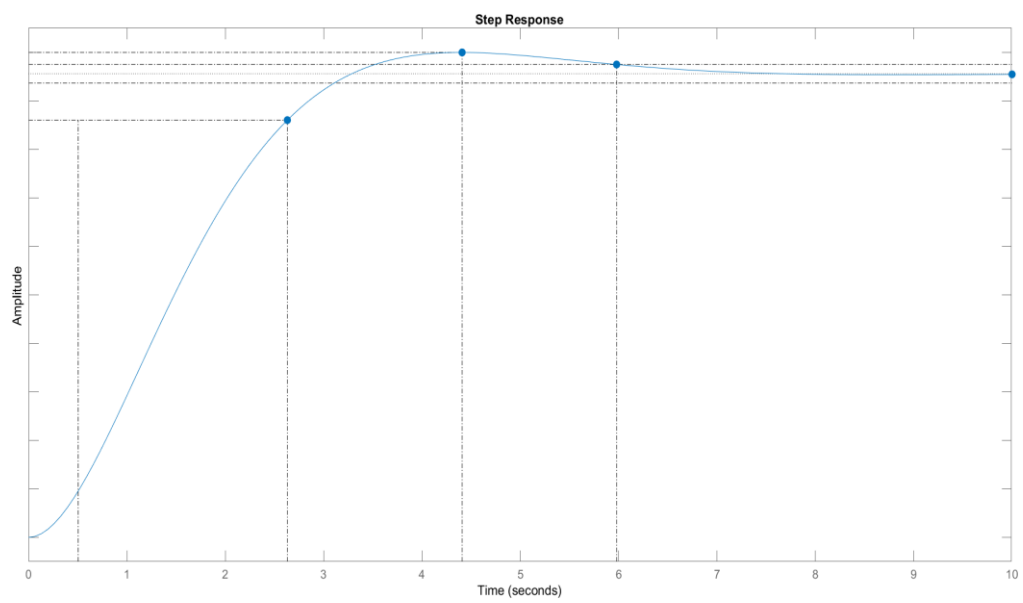
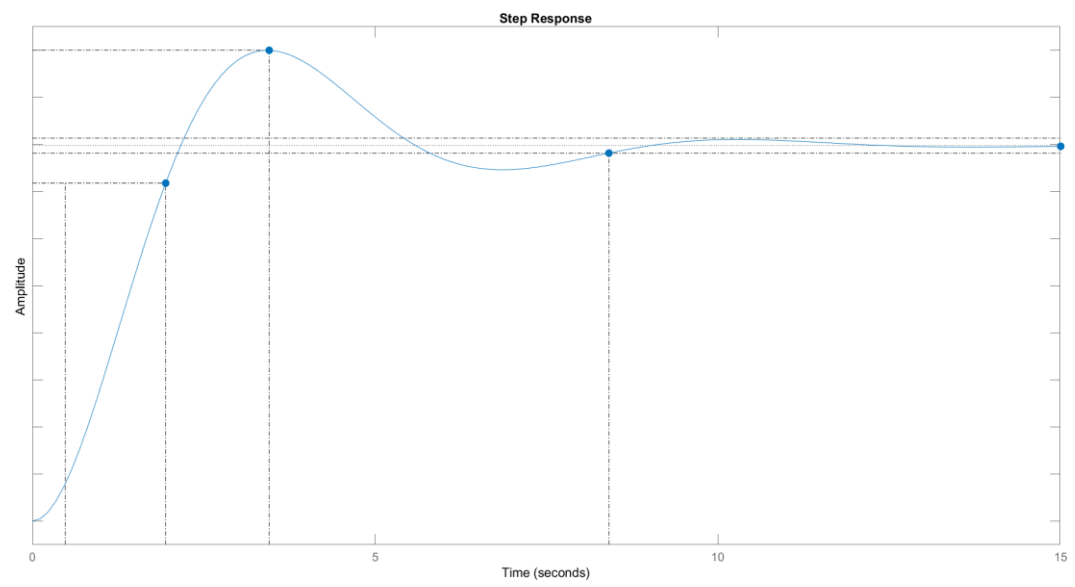
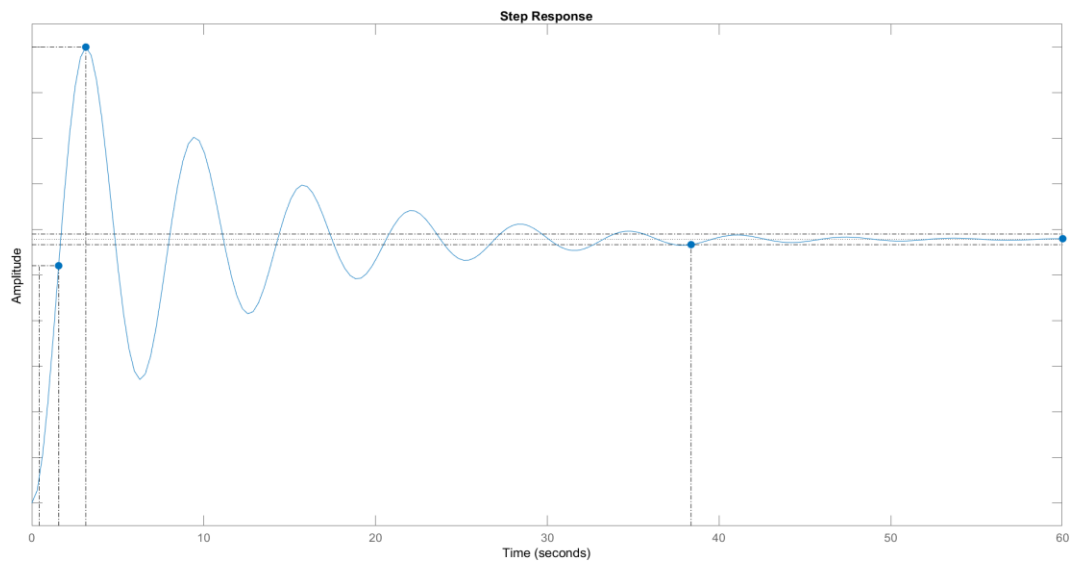
Percentage overshoot shows how much the response goes above the final value before settling down. It is calculated by:  $100e^{-\zeta\pi/\sqrt{1-\zeta^2}}$

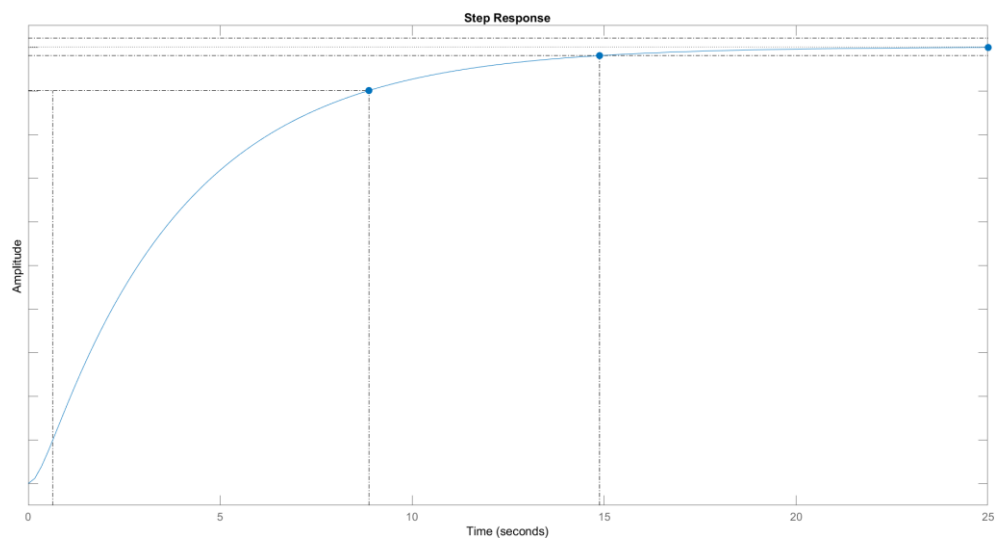
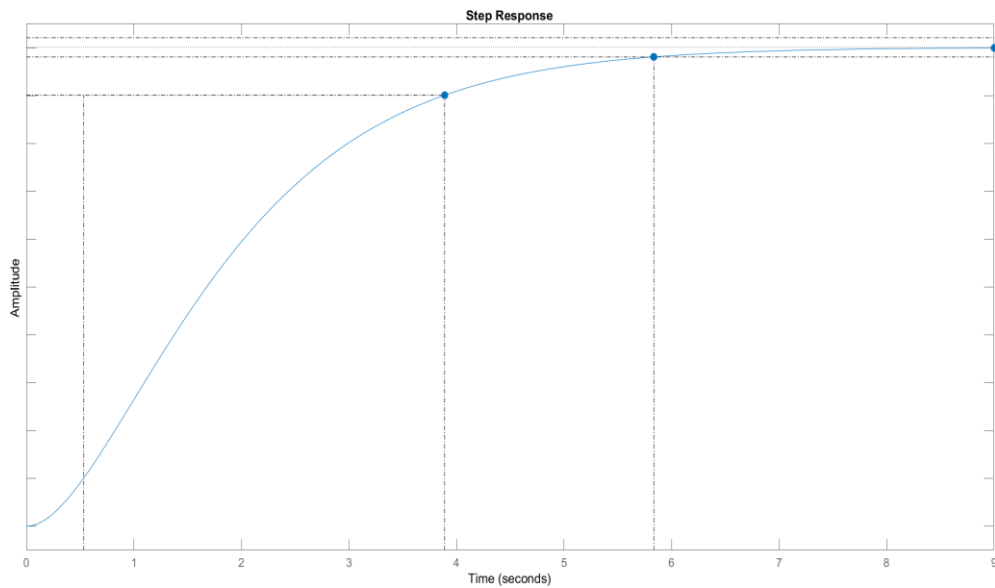
If the percentage overshoot is high, it usually means the damping ratio is low, and the system has more oscillations.



For this system, where  $\omega_n = 1$ , we should define rise time, settling time, and percentage overshoot for different values of  $\zeta$ .

| $\zeta$ | Rise time | Settling time | Percentage overshoot |
|---------|-----------|---------------|----------------------|
| 0.1     | 1.13      | 38.38         | 72.91 %              |
| 0.4     | 1.46      | 8.4           | 25.37 %              |
| 0.7     | 2.13      | 5.98          | 4.6 %                |
| 1       | 3.36      | 5.83          | 0 %                  |
| 2       | 8.23      | 14.88         | 0 %                  |





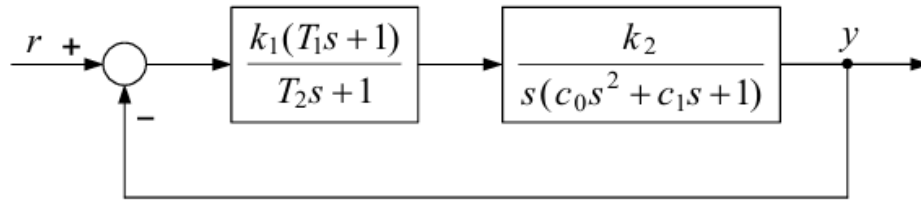
## Discussion of the results obtained.

We can see from the table and the graphs that when the damping ratio increases, the rise time becomes longer, but the settling time and percentage overshoot become smaller.

However, when the system is overdamped ( $\zeta = 2$ ), the rise time and settling time increase a lot. This happens because too much damping makes the system move very slowly. It does not overshoot the final value, but it takes more time to reach and stay at that value.

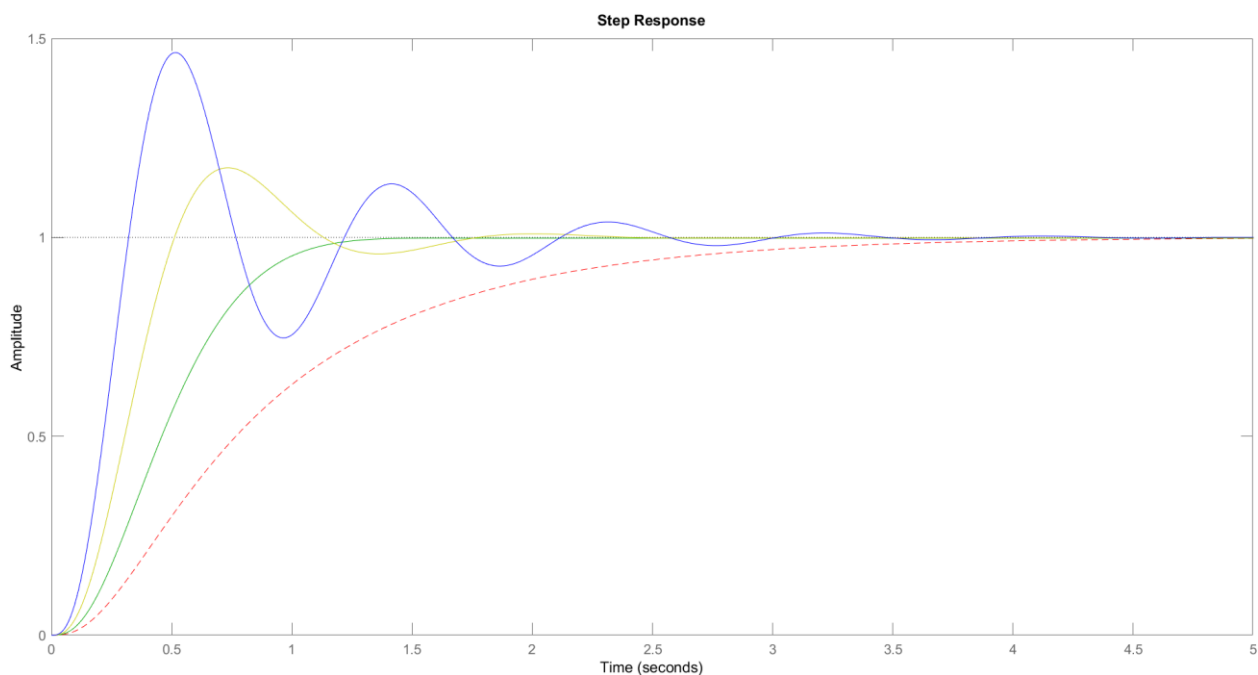
## Task 4: Parametric Control System Analysis

Consider the control system model shown in Figure 5. The values of the parameters are  $T_1 = 0.25$  s,  $T_2 = 0.125$  s,  $k_2 = 0.4$ ,  $c_0 = 0.02$  s<sup>2</sup>,  $c_1 = 0.3$  s.



This task involves the analysis of the temperature control system model shown in Figure. The parameters are fixed at  $T_1=0.25$ ,  $T_2=0.125$ ,  $k_2=0.4$ ,  $c_0=0.02$ , and  $c_1=0.3$ . The system is tested for different controller gains  $k_1$  in  $\{2.5, 5, 10, 20\}$ .

| $k$ | Rise time | Settling time | Percentage overshoot |
|-----|-----------|---------------|----------------------|
| 2.5 | 1.78      | 3.34          | 0 %                  |
| 5   | 0.67      | 1.13          | 0 %                  |
| 10  | 0.32      | 1.6           | 17.5 %               |
| 20  | 0.2       | 2.8           | 46.46 %              |



## Discussion of the results obtained.

When the value of  $k$  increases, the system becomes more damped. For  $k = 2.5$  and  $k = 5$ , the system is underdamped and the percentage overshoot is 0. For  $k = 10$  and  $k = 20$ , the system becomes overdamped, and we can notice an increase in settling time and percentage overshoot.

In conclusion, for underdamped systems, even if the overshoot is 0, the response is quite slow. For overdamped systems, the response is faster and stronger, but higher percentage overshoot can be seen.

## CONCLUSION

The laboratory exercises successfully demonstrated the practical application of time-domain performance analysis. Through Task 3, the influence of the damping ratio on overshoot and settling time was confirmed. Task 4 highlighted the trade-off between controller gain and system stability. MATLAB proved to be an essential tool for visualizing these dynamic behaviors and determining precise performance metrics.